

1 **Title page**

2 **A Prediction-Based Framework for Developing Integrated Pedestrian Safety Systems**
3 **under High-Risk Scenarios**

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14 **Highlights**

- 15 1. A prediction-based framework using a vehicle-view HITL dataset addresses reaction-time bottlenecks in high-
16 risk interactions.
- 17 2. The Causal CNN-LSTM model achieves sub-meter accuracy (FDE=0.22 m) in capturing abrupt maneuvers
18 during unavoidable collisions.
- 19 3. Prediction-based triggering secures a 200 ms lead time, ensuring full active hood deployment before impact.
- 20 4. Shifting to pre-impact prediction reduces the Head Injury Criterion (HIC) by 62.1%, bridging active sensing and
21 passive safety.

22

ABSTRACT

Pedestrians are among the most vulnerable road users and remain a primary focus of intelligent vehicle safety. In high-risk scenarios where avoidance is physically infeasible, vehicles' front-end structural designs (e.g., active hoods) play a crucial role in injury mitigation. However, existing systems typically rely on contact-based sensors, which leave extremely short time for device deployment. To address this challenge, this study proposes a prediction-based integrated pedestrian safety framework. We constructed a vehicle-perspective, human-in-the-loop virtual reality dataset specifically enriched with high-risk boundary cases (near-misses and collisions). Using this data, we developed a short-horizon forecasting model (combining causal CNN and stacked LSTM), which achieved robust sub-meter accuracy in capturing abrupt pedestrian behavior. Compared with conventional contact-based systems, the proposed prediction-based triggering method advances the actuation timing by 200 ms. This gives the active hood actuator more time margin to ensure it fully deploys before the pedestrian's head makes contact. Integrated simulation results demonstrated that this anticipation enabled the active hood to fully deploy before head contact, thereby realizing a 62.1% reduction in head injury severity. Further validation on real-world crash accidents confirmed the framework's robustness and transferability. Overall, this methodological framework supports a paradigm shift from post-impact detection to pre-impact prediction, enabling the next generation of proactive pedestrian protection systems.

Keywords: Pedestrian safety, Intelligent vehicles, Prediction-based trigger approach, Human-in-the-loop, Collision mitigation

1. Introduction

Vulnerable road users (VRUs), particularly pedestrians, account for a disproportionately high share of fatalities and severe injuries in road traffic accidents worldwide. Specifically, pedestrians alone comprise approximately 23% of the estimated 1.19 million annual road traffic deaths (WHO 2023). Specifically, fatalities and severe injuries frequently occur when a pedestrian's head impacts the vehicle's stiffer underlying structures, such as the engine block or the windshield frame. With the rapid development of highly automated vehicles (HAVs), ensuring pedestrian protection has become a critical component of intelligent transportation systems.

High-risk scenarios—situations where collisions are extremely difficult to avoid—pose special challenges for pedestrian protection. While automated vehicles (AVs) are designed to maximize safety, a substantial portion of real-

world crashes remain unavoidable due to unpredictable pedestrian behaviors (e.g., sudden intrusions) or limited physical space for evasive action (Bosia et al. 2024, Sun et al. 2024, Zhou et al. 2024). In such cases, where collision avoidance becomes infeasible, the safety focus should pivot from avoidance to injury mitigation (Bosia et al. 2024, Li et al. 2022). In these unavoidable collision scenarios, time is the governing constraint. Studies indicate that pedestrian head impact typically occurs 60–120 ms after the initial bumper contact (Crocetta et al. 2015, Hu, Monfort and Cicchino 2024, Yin, Li and Xu 2017). The reliance of traditional pedestrian protection systems on contact-based sensors severely restricts the development of advanced countermeasures, such as pedestrian airbags or variable-position bumpers, due to the insufficient deployment window post-contact (Li et al. 2024). Currently, only pyrotechnic active hoods can actuate rapidly enough to be effective within this tight timeframe; however, pyrotechnic mechanisms significantly increase lifecycle costs and lack reversibility (Lee et al. 2017). Consequently, extending the reaction window by even 100–200 ms via pre-impact prediction is critical. This extension offers dual benefits: it unlocks the feasibility of a broader range of protective devices, and it enhances the economic viability and practicality of existing systems—for instance, by enabling the replacement of single-use pyrotechnics with cost-effective, reversible electric actuators (Xu et al. 2010).

A natural remedy to these timing constraints is to earn time before impact. If an onboard system can accurately predict that a crash is imminent—and quantify the expected impact conditions—then protection mechanisms can be initiated earlier. This pre-emptive action secures the milliseconds necessary for reliable deployment and stiffness modulation, effectively bridging the gap identified above. While traditional vehicles struggle to achieve this, intelligent vehicles equipped with rich perception suites (e.g., LiDAR, cameras) and onboard inference capabilities make such pre-impact forecasting increasingly feasible (Parseh et al. 2019). The challenge is reliable accuracy under severe time pressure. Off-the-shelf models trained on public corpora—where high-risk episodes are long-tailed—degrade as the window shrinks; motion becomes more stochastic/non-smooth, and learned priors fail where pre-impact fidelity is most needed (Anderson et al. 2019, Chen et al. 2024). Hence the primary bottleneck is data, not architecture: we require a vehicle-view, pre-impact dataset densely capturing near-misses and collisions with synchronized sensing and outcome ground truth. Only with such data can a short-horizon predictor be trained/validated to retain accuracy under high time pressure; otherwise, architectural tweaks regress to average-case behavior rather than impending impact conditions (Wu et al. 2020).

Overall, two major gaps hinder the development of prediction-based trigger pedestrian protection systems for automated vehicles. First is the data bottleneck. High-risk, pre-crash datasets that are both sufficiently large and information-complete remain scarce (Campolettano, Scanlon and Victor 2023, Zhou et al. 2023). What is needed are datasets encompassing near-misses and crash scenarios with synchronized vehicle-centric perception, pedestrian kinematics/scene context and injury-relevant outcome proxies (Wang et al. 2024). However, in-field data collection is constrained by the rarity of crashes and the burden of instrumentation; simulation and human-in-the-loop technologies provide partial relief but remain unevenly applied and insufficiently standardized for injury-oriented design requirements (Liu et al. 2025).

Second is the model bottleneck. Existing single-method predictors—typically relying on trajectories or single sensing modalities—struggle to capture abrupt pedestrian maneuvers (e.g., sudden stop-and-go) and impending collision dynamics under severe time pressure (Alghodhaifi and Lakshmanan 2023, Campolettano et al. 2023). These models seldom deliver calibrated pre-impact predictions of collision conditions (relative position/velocity, time-to-impact) within the tight tens-of-milliseconds budget, nor can they map such predictions to actionable injury-risk estimates for control (Li et al. 2021). These unresolved gaps motivate the need for an integrated, prediction-based trigger pathway for pedestrian protection.

To address these challenges, this paper proposes a prediction-based framework for developing integrated pedestrian safety systems under high-risk scenarios. The present work adopts the human-in-the-loop (HITL) virtual reality protocol (Li et al. 2025) to further address a distinct engineering question: integrating onboard sensing, short-horizon forecasting, and pre-impact intervention within an automotive safety pipeline. Specifically, our main contributions are threefold:

1. **Data:** we constructed a vehicle-view HITL dataset purpose-built for hazardous scenarios. It captures 1,920 episodes (including 618 collisions) with synchronized vehicle perception and ground-truth outcome labels, resolving the data bottleneck for injury-oriented learning.
2. **Model:** leveraging the high-risk kinematic features from this corpus, we developed a short-horizon trajectory forecasting model (combining causal CNN and stacked LSTM). Unlike baselines that degrade under tight time pressure, this model robustly captures abrupt pedestrian maneuvers, achieving accurate collision-condition predictions without relying on future information leakage.

3. Framework validation: building on the precise short-horizon forecasts, we designed a complete semantic-gated, prediction-based trigger actuation pipeline with its application to an active hood. Integrated simulation results demonstrate that this pipeline expands the effective reaction window from approximately 50 ms to 200–250 ms. By ensuring successful deployment where reactive systems might fail due to time constraints, our approach achieves a 62.1% reduction in the Head Injury Criterion (HIC) relative to a standard fixed hood baseline. Furthermore, the trained forecasting model was validated on real-world crash cases to verify its robustness and transferability (Figure 1).

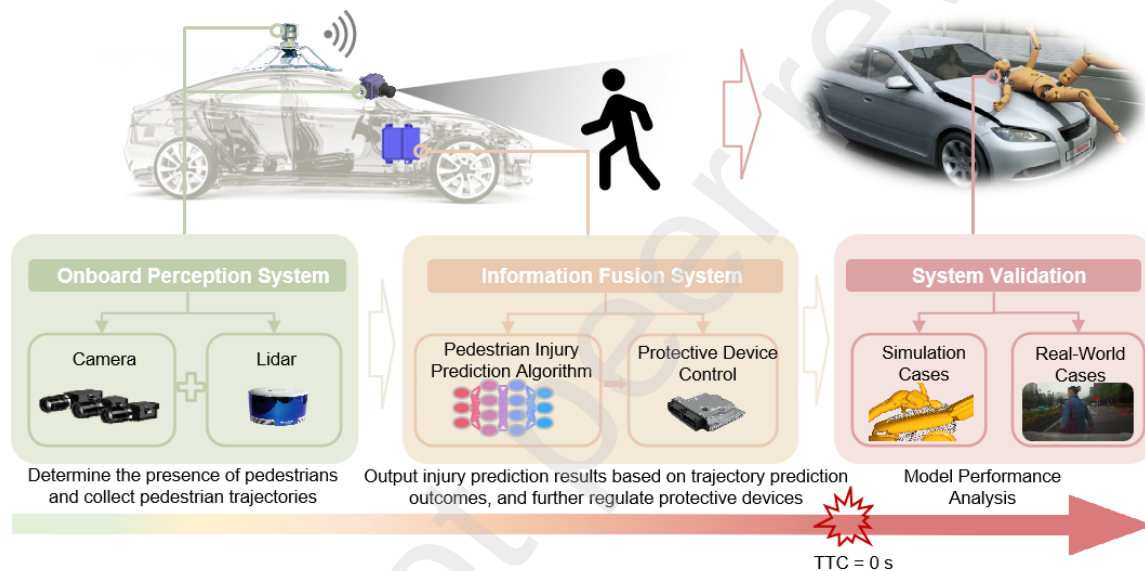


Fig. 1 Integrated prediction-based trigger pedestrian protection framework

2. Methods

2.1 Construction of Vehicle-View High-Risk Dataset via Human-in-the-Loop Experiments

We constructed a vehicle-view dataset for high-risk pedestrian–vehicle interactions by integrating a human-in-the-loop (HITL) VR environment with synchronized CARLA simulation (Figure 2). Participants performed time-pressured road-crossing tasks to elicit natural reactive behaviors and abrupt maneuvers. We defined the vehicle coordinate system with the origin at the center of the front bumper, the X-axis pointing forward (longitudinal), and the Y-axis pointing left (lateral).

While the HITL protocol (participants, arena layout, and task instructions) followed our previous work, this study introduced specific automotive-oriented extensions: (i) synchronized vehicle-side logging; (ii) pre-impact trigger logic; and (iii) kinematic exports for injury modeling. The experimental design employed a full-factorial structure (64

archetypes, detailed in Table 1), varying ego-vehicle speeds (15–45 km/h), interaction zones, crossing directions, and occlusion levels. This design ensured the dataset covered the transition from deliberative crossings to near-reflexive evasive actions. In total, 1,920 episodes were recorded, of which 618 (32.2%) resulted in collisions, providing a balanced corpus for high-risk learning.

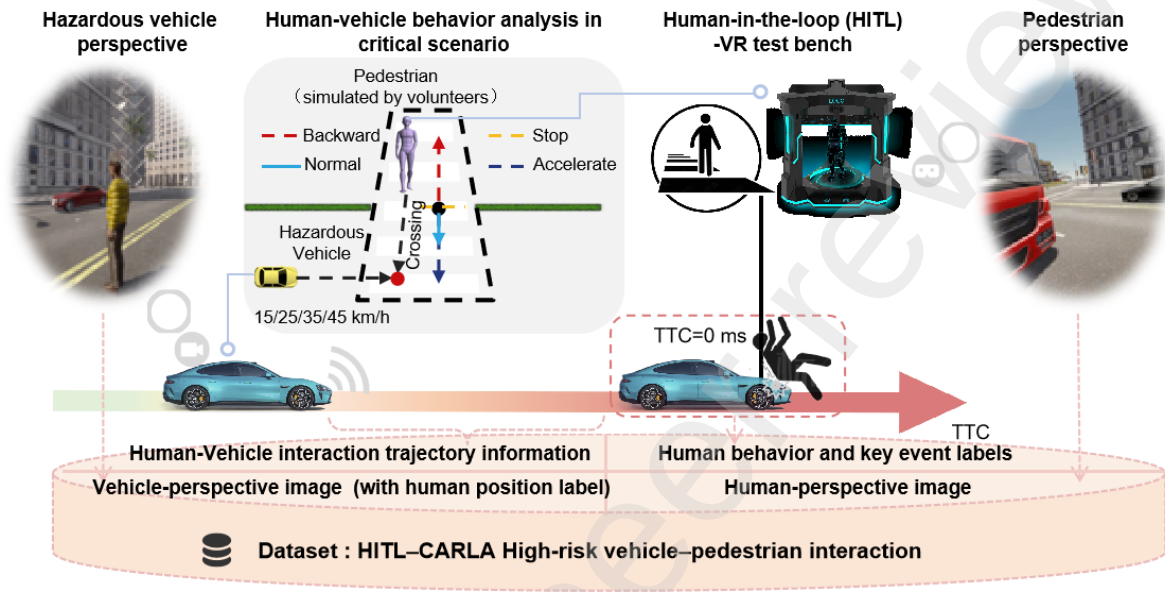


Fig. 2 Human-in-the-loop to CARLA pipeline for high-risk dataset generation

Scenario coverage followed a full-factorial design yielding 64 archetypes: 2 scenario types (mid-block vs. unsignalized crosswalk) $\times$ 4 ego-speed layers (15/25/35/45 km/h) $\times$ 2 interaction zones (near-side vs. far-side relative to the ego lane) $\times$ 2 approach directions (left-to-right vs. right-to-left) $\times$ 2 occlusion conditions (with vs. without static/slow obstacles). The four speeds span progressively tighter time budgets—i.e., shorter time-to-contact at first detection as speed increases—so the corpus covers the transition from deliberative to near-reflexive interactions. Near/far zones encode coarse spatial relations relevant to lateral risk; bidirectional approaches remove directional priors; and the occlusion factor introduces line-of-sight or path-of-travel restrictions. Across all combinations, each archetype is repeated by distinct participants (30 per archetype), producing 1,920 episodes in total, of which 618 culminate in vehicle–pedestrian contact and the remainder are high-risk near-misses.

Table 1. Factor levels for the HITL–CARLA high-risk vehicle–pedestrian design

Factor	Levels	Operational notes
Scene type	Mid-block; Crosswalk	Two canonical pedestrian–vehicle interaction scenarios used in the HITL study; reconstructed one-to-one in CARLA from the vehicle view.
Vehicle speed	15; 25; 35; 45 km/h	Discrete approach speeds to control time-pressure (time to contact (TTC)) exposure.

Interaction zone	Near; Far	Crossing starts on the near-side vs far-side relative to the ego vehicle's lane.
Crossing direction	L→R; R→L	Pedestrian moves from the left to right (L→R) or right to left (R→L) in the ego-vehicle frame.
Occlusion	With; Without	Presence/absence of an obstacle vehicle blocking line-of-sight (occluded vs non-occluded).

Notes: Full factorial design yields 64 archetypes (2 scenes $\times$ 4 speeds $\times$ 2 zones $\times$ 2 directions $\times$ 2 occlusion levels); 1,920 episodes in total, including 618 contacts (32.19%). Sampling step $\Delta t = 0.1$ s. Abbreviations: TTC, time-to-collision; HIC, Head Injury Criterion.

Table 1 enumerates the factor levels and operational notes that define the exposure space, clarifying how speed layers shape the time budget and how occlusion and approach geometry concentrate collision risk across archetypes.

Each VR episode was then reconstructed from the ego-vehicle perspective in CARLA, preserving geometry and timing so that we could generate vehicle-view sensing under identical conditions (see Figure 3). The simulated sensor rig included a forward camera (vehicle-view imagery) and light detection and ranging (LiDAR) (point clouds used to derive precise ego-pedestrian relative geometry in the vehicle frame). These streams served complementary training purposes within this dataset: trajectory prediction consumed the LiDAR-derived relative positions (and associated kinematic states) to learn short-horizon pedestrian motion relative to the vehicle, while image detection used camera frames that are auto-labeled via a calibrated 3D-to-2D projection from CARLA ground-truth actor states to produce image-plane pedestrian boxes/IDs without manual annotation.

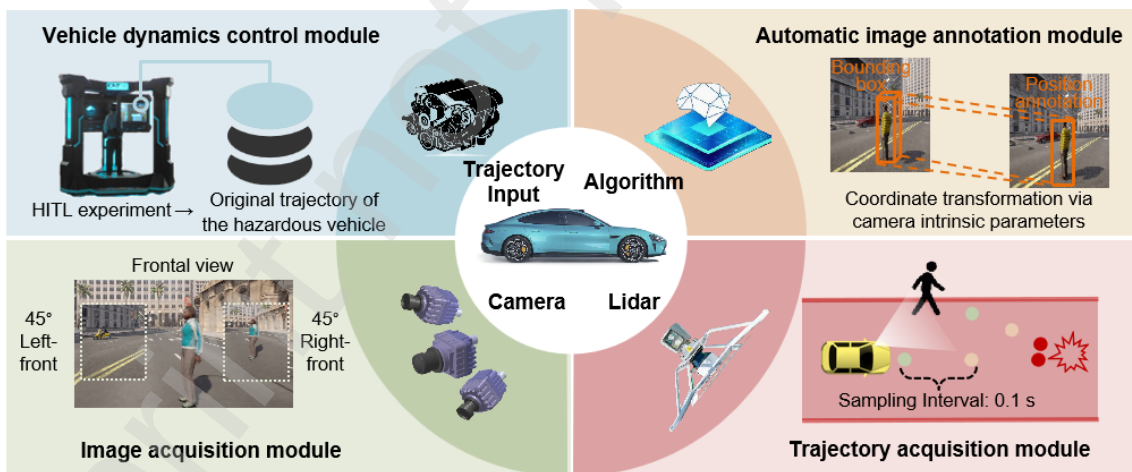


Fig. 3 Vehicle-view dataset assembly and sensing interfaces

All streams were synchronized at a fixed 0.1 s sampling step to match common automotive sensing rates and to support the short-horizon learning used later. Each episode package contains: (i) pedestrian states (x, y, z), speed, and heading; (ii) ego-vehicle states (position, speed, yaw angle; additional kinematics such as yaw rate and

longitudinal/lateral accelerations can be computed offline); (iii) vehicle-view camera frames and LiDAR point clouds; and (iv) behavior tags and event markers. Table 2 provides a complete list of signal fields—definitions, units, and sampling rates—thereby making the sensing–prediction–control interfaces auditable and supporting reproducible training and downstream controller mapping. Descriptive statistics by archetype and exposure (counts, collision share, distributions of initial separations, ego speed, and time-to-contact at first detection) are reported in Section 3.1 to avoid duplication here.

Table 2. Vehicle-view data streams and key fields

Stream	Field (symbol)	Definition	Unit / rate	Source	Used for
Camera	Image frame	Forward vehicle-view RGB image	px; $\Delta t = 0.1$ s	CARLA camera	Detection (real-time detection transformer (RT-DETR))
Camera	2D pedestrian box	Auto-labeled image-plane bounding box via calibrated 3D→2D projection	px	CARLA metadata → 3D→2D	Detection training & eval
Camera	Detection confidence (conf)	RT-DETR human class posterior	[0,1]; per frame	RT-DETR	Semantic gate in control
LiDAR	Relative position	Ego-centric pedestrian position (vehicle frame)	m; $\Delta t = 0.1$ s	CARLA LiDAR → tracking	Forecast input; control geometry
LiDAR	Relative speed v_{rel}	Ego-centric relative speed (finite-difference or tracker output)	m/s	Derived from p_{rel}	Risk features; control
Ego state	Vehicle speed $p_{rel} = (\Delta X, \Delta Y, \Delta Z)$	Longitudinal ego speed	km/h	CARLA vehicle state	Valid speed window for actuation
Ego state	Yaw angle	Vehicle heading (for frame alignment)	deg	CARLA vehicle state	Frame alignment & QC
Events	Timestamps (spawn, decision change, conflict entry, first detection, anchor)	Episode key instants; anchor = contact (collision) or closest approach (near-miss)	s	HITL log + CARLA	Timing alignment; TTC metrics
Factors	Scenario / speed / interaction zone / direction / occlusion	Factor levels used in 64 archetypes	categorical	Protocol	Stratified analysis / splits
IDs	subject_id, episode_id	Participant and trial identifiers	—	Protocol	Subject-wise split (train/val/test)
Labels	Behavior tags	Commit/abort, stop/accelerate, turn, etc.	categorical	HITL annotation	Stratification; qualitative cases

Notes: Sampling step $\Delta t = 0.1$ s across streams. Camera 2D boxes are auto-labeled via CARLA 3D→2D projection; no manual annotation. The forecasting module consumes LiDAR-derived p_{rel} only; the controller operates on the synchronized triplet {detection confidence, current p_{rel} , predicted +0.2 s condition}.

2.2 Short-Horizon Trajectory Forecasting based on Causal CNN-LSTM Architecture

Using the vehicle-view dataset described in Section 2.1 (camera images with auto-labels and LiDAR-derived ego–pedestrian geometry), we designed a forecasting module that is strictly sequence-to-sequence on relative position.

Concretely, at each decision instant the model ingested a 4-frame history of ego-pedestrian relative positions sampled at a fixed step $\Delta t = 0.1$ s (consistent with common automotive sensing rates)(Alemneh, Senouci and Brunet 2019), and output a 2-frame forecast at +0.1 s and +0.2 s. Thus the input window spanned 0.4 s and the prediction horizon spanned 0.2 s.

$$[p_{rel}(t-0.3), p_{rel}(t-0.2), p_{rel}(t-0.1), p_{rel}(t)] \xrightarrow{f_{\theta}} [\hat{p}_{rel}(t+0.1), \hat{p}_{rel}(t+0.2)] \quad (1)$$

where $p_{rel}(t) \in \mathbb{R}^2$ is the ego-frame relative position vector at time t , i.e., $p_{rel}(t) = [\Delta X(t), \Delta Y(t)]^T$ (meters); the hat $\hat{\cdot}$ denotes a prediction; f_{θ} is the causal forecaster (CNN+LSTM) with parameters θ that consumes a 4-step history sampled at interval Δt (typically 0.1 s) without future leakage; all vectors are column vectors; axis convention: $+\Delta X$ forward (longitudinal), $+\Delta Y$ left (lateral).

We employed a CNN+LSTM fusion architecture (see Figure 4). A causal 1-D temporal CNN stem encoded the 4-step history of relative positions, extracting short-horizon motion signatures without future leakage and providing a compact latent sequence. On top of this stem, a stacked LSTM captured nonlinear maneuver transitions characteristic of high-risk interactions (e.g., sudden acceleration/deceleration, quick re-orientation). This division of labor is intentional: the CNN stabilizes local kinematics and temporal causality, while the LSTM retains longer context and improves sensitivity to abrupt intent shifts. Inputs were standardized per axis; lightweight rate cues (e.g., first-order differences) were learned internally by temporal convolutions, keeping the interface purely positional and device-agnostic.

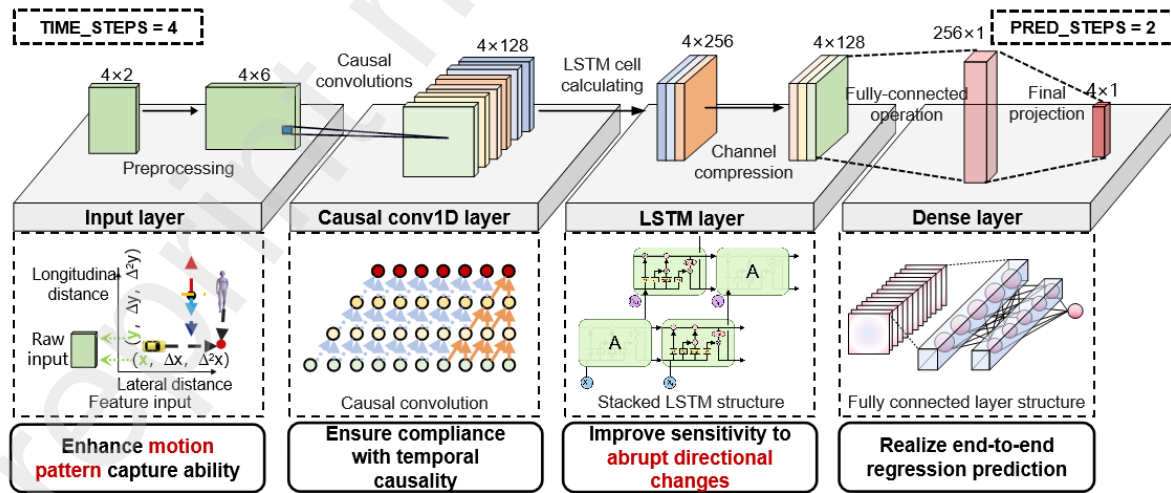


Fig. 4 CNN+LSTM forecaster architecture

From the two forecasts, we estimated the relative impact speed at $t+0.2$ s by forward differencing:

$$\hat{v}_{rel}(t + 0.2) = \frac{\hat{p}_{rel}(t + 0.2) - \hat{p}_{rel}(t + 0.1)}{\Delta t} \quad (2)$$

where $\hat{v}_{rel}(t + 0.2) \in \mathbb{R}^2$ is the ego-frame relative velocity at $t + 0.2$ s estimated by the forward difference of predicted positions; Δt is the interval between $\hat{p}_{rel}(t + 0.1)$ and $\hat{p}_{rel}(t + 0.2)$ (e.g., 0.1s); components $\hat{v}_{rel} = [\hat{V}_x, \hat{V}_y]^\top$ are in m/s.

A collision gate determined whether the $t+0.2$ s position qualified as a predicted collision position. Contact was declared only when the forecast lay within calibrated proximity thresholds δ_x, δ_y (meters) around the vehicle front, optionally with a simple time-consistency check:

$$\hat{c}(t + 0.2) = \mathbf{1}(|\hat{\Delta X}(t + 0.2)| \leq \delta_x \wedge |\hat{\Delta Y}(t + 0.2)| \leq \delta_y) \quad (3)$$

where $\mathbf{1}(\cdot)$ is the indicator function (returns 1 if the condition holds and 0 otherwise); $\hat{\Delta X}(t + 0.2)$, $\hat{\Delta Y}(t + 0.2)$ are the predicted longitudinal/lateral displacements at $t + 0.2$ s (meters); $\delta_x, \delta_y > 0$ (meters) specify the danger envelope thresholds along the two axes; $\wedge$ denotes logical AND. Hence $\hat{c}(t + 0.2) = 1$ indicates the forecast enters the calibrated envelope used by the trigger logic.

If $\hat{c}(t + 0.2) = 1$, $\hat{p}_{rel}(t + 0.2)$ is treated as the predicted collision position and $\hat{v}_{rel}(t + 0.2)$ as the predicted impact speed; otherwise, the $+0.2$ s output is used as a non-colliding relative position for risk assessment rather than impact tagging. Thresholds δ_x, δ_y were calibrated on the development split from contact geometry (vehicle front plane and pedestrian width) or empirical percentiles of $(|\Delta X|, |\Delta Y|)$ at contact. (If needed, TTC can be derived downstream from the predicted geometry/velocity; the control layer uses these quantities directly.

Training was cast as supervised regression at the two future horizons with a robust smooth-L1 (Huber) penalty on axis-wise errors, using axis weights to reflect longitudinal/lateral uncertainty asymmetry and a horizon weight to emphasize the $+0.2$ s control target. Optimization followed standard practice (Nadam, initial LR $\approx 1e-3$; Reduce-on-Plateau; early stopping; light dropout on LSTM hidden states). To avoid leakage and assess generalization across individuals, we adopted a subject-wise split (e.g., 30 participants $\rightarrow$ 20/5/5 for train/validation/test), while balancing scenario types and occlusion layouts across partitions. For ablation, three predictors shared identical I/O and protocols: LSTM-only, single-layer CNN-LSTM, and two-layer CNN-LSTM (full model). Comparative errors (ADE/FDE at $+0.1/+0.2$ s and axis-wise root-mean-square error (RMSE)) are reported in Results. The detection branch (RT-DETR

on camera images with auto-label) was trained separately; the forecasting module uses LiDAR-derived relative positions, preserving a clean module boundary for the application-level timing and control described later.

$$\mathcal{L} = \sum_{m \in \{0.1, 0.2\}} \beta_m [w_x \rho(\hat{\Delta X}(t+m) - \Delta X(t+m)) + w_y \rho(\hat{\Delta Y}(t+m) - \Delta Y(t+m))] \quad (4)$$

where $\rho(\cdot)$ is the Huber (smooth-L1) loss with scale parameter δ (default $\delta = 1$ unless stated); $w_x, w_y > 0$ weight axis-wise errors; β_m are horizon weights with $\beta_{0.1} = 1$ and $\beta_{0.2} = \alpha \geq 1$ to emphasize the +0.2 s horizon used by control; $\hat{\Delta X}(t+m)$, $\hat{\Delta Y}(t+m)$ and $\Delta X(t+m)$, $\Delta Y(t+m)$ are predicted and ground-truth displacements (meters); $m \in \{0.1, 0.2\}$ s. If mini-batch training is used, $\mathcal{L}$ is averaged over the batch.

2.3 Semantic-Gated Trigger Policy and Integrated Safety Framework Validation

Building upon the vehicle-view dataset and the short-horizon forecasting module, we integrated perception, relative-state sensing, and actuation decision-making into a unified safety pipeline. In this pipeline, camera imagery was processed by RT-DETR to generate pedestrian hypotheses (Zhao et al. 2024), while synchronized LiDAR data supplied precise vehicle-frame relative geometry at 0.1 s intervals. The central controller operated on a synchronized triplet—vision confidence, current geometry, and the predicted impact condition at $H = 0.20$ s—and adhered to a prediction-first triggering principle. Specifically, the system commanded actuation only when the predicted position fell within a calibrated danger envelope (set to $|y| \leq 0.8$ m and $x < 0.5$ m to prioritize safety coverage) and the estimated impact speed exceeded the device threshold (benchmarked at 25 km/h). To minimize false activations from non-human objects, a semantic gate was implemented, requiring both a calibrated confidence threshold and 3D-2D association consistency; only tracks satisfying these strict criteria were deemed eligible for pre-impact intervention.

To quantify the potential for injury mitigation, we instantiated this control policy on an active hood system (AHS) within a high-fidelity simulation environment. We designed a three-arm comparative study to isolate the benefits of timing logic from hardware parameters: (i) a fixed hood (baseline), (ii) a collision-triggered hood (reactive), and (iii) the proposed prediction-based trigger hood (proactive). Vehicle – pedestrian interaction trajectories were sourced from the CARLA dataset, and the resulting trigger commands and impact poses were fed into a MADYMO multibody solver. Specifically, we employed the validated vehicle-pedestrian impact model to ensure kinematic fidelity (Li et al. 2024). This setup allowed us to simulate pedestrian kinematics and compute the Head Injury Criterion (HIC) under

identical crash conditions, thereby directly attributing any safety gains to the extended lead time provided by the forecasting module.

Finally, to assess the framework's generalizability beyond the synthetic environment, we evaluated the simulation-trained forecaster on 65 real-world crash clips recorded by onboard sensors. In this offline validation, no physical actuation was simulated; instead, the study focused on the model's ability to accurately predict collision conditions purely based on real-world camera and LiDAR logs. By applying the same forecasting logic to these naturalistic episodes, we verified that the system's ability to capture abrupt maneuvers and lock onto impact targets persisted in real-world scenarios, validating the robustness of the proposed approach.

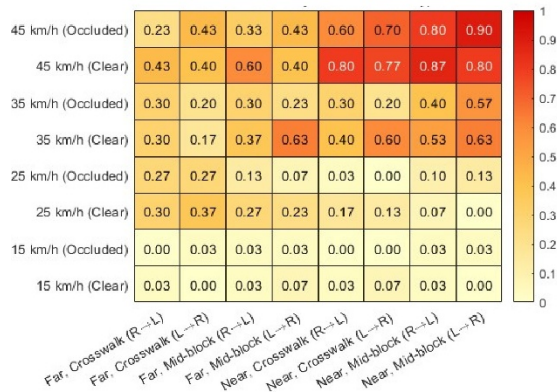
3. Results

3.1 Exposure characteristics of the high-risk dataset

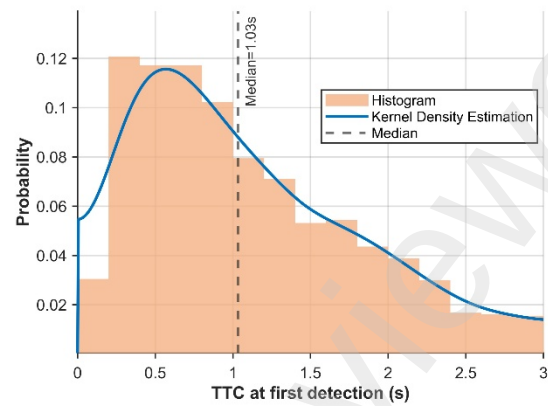
The dataset was constructed to cover safety-critical regimes involving near-misses and physical contacts. A substantial proportion of the recorded episodes (32.2%) resulted in actual vehicle – pedestrian impacts.

Figure 5(a) illustrates the collision share across the 64 scenario archetypes. The rows represent combinations of ego-vehicle speed (15–45 km/h) and visibility conditions (Clear vs. Occluded). The columns classify pedestrian behavior by origin (Far-side vs. Near-side), crossing environment (Crosswalk vs. Mid-block), and walking direction. High collision frequencies were concentrated in conditions characterized by higher speeds ($v \geq 35$ km/h), partial occlusions, and near-side intrusions. In these archetypes, the pedestrian appeared from a short lateral offset directly into the vehicle's path, leaving minimal physical margin for avoidance.

Figure 5(b) presents the distribution of Time-to-Collision (TTC) at the moment of first stable detection. The distribution peaked below 1.0 s, with a significant fraction of cases having a TTC of less than 0.5 s at the initial detection instance. This indicated that the dataset captured unavoidable or near-limit encounters where the available reaction window was significantly compressed compared to typical long-range detection scenarios.



(a) Per-archetype collision share heatmap



(b) Dataset-level TTC distribution

Fig. 5 Dataset-level exposure characterization: collision share by scenario archetype (a) and first-detection TTC distribution (b)

3.2 Forecasting accuracy

We evaluated the forecasting module's ability to predict the ego-frame pedestrian position at short horizons ($H = +0.1s$ and $+0.2s$). Performance was quantified using standard metrics: Final Displacement Error (FDE), defined as the Euclidean distance between the forecasted and ground-truth position at horizon H ; Average Displacement Error (ADE), the mean distance over the prediction steps; and axis-wise Root-Mean-Square Error (RMSE).

Table 3 summarizes the performance on the held-out evaluation split. At the control-critical horizon ($H = +0.2s$), the proposed model achieved a median FDE of 0.220 m and ADE of 0.242 m. These errors were consistently lower than those of the LSTM-only baseline (FDE 0.620 m, ADE 0.674 m) and the shallow CNN - LSTM baseline (FDE 0.708 m, ADE 0.487 m), and significantly outperformed the constant-velocity "Hold" baseline (FDE 1.633 m). Axis-wise analysis showed similar trends, with our model achieving the lowest longitudinal and lateral RMSE (0.227 m and 0.208 m, respectively).

Table 3. Performance across forecast horizons for relative vehicle-pedestrian position trajectory prediction

H (s)	Model	n (cases)	FDE, median [IQR] (m)	ADE, median [IQR] (m)	RMSE_x, mean (SD) (m)	RMSE_y, mean (SD) (m)
0.100	hold	245	0.847 [0.536]	0.968 [0.414]	0.797 (0.299)	0.407 (0.161)
0.100	Shallow CNN+LSTM	245	0.419 [0.260]	0.376 [0.203]	0.257 (0.130)	0.419 (0.744)
0.100	LSTM-only	245	0.505 [0.315]	0.496 [0.337]	0.358 (0.252)	0.617 (0.968)
0.100	Ours	245	0.145 [0.144]	0.165 [0.096]	0.164 (0.135)	0.127 (0.146)
0.200	hold	245	1.633 [0.944]	1.931 [0.750]	1.548 (0.567)	0.834 (0.381)
0.200	Shallow CNN+LSTM	245	0.708 [0.374]	0.487 [0.267]	0.308 (0.168)	0.533 (0.682)
0.200	LSTM-only	245	0.620 [0.380]	0.674 [0.405]	0.414 (0.250)	0.785 (1.289)

0.200	Ours	245	0.220 [0.245]	0.242 [0.139]	0.227 (0.169)	0.208 (0.304)
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Notes: H = forecast horizon (seconds). FDE = final displacement error at H. ADE = average displacement error over $[0, H]$. $RMSE_x / RMSE_y$ = axis-wise root-mean-square error along vehicle longitudinal (x) and lateral (y) axes. IQR = interquartile range; SD = standard deviation. Units are meters; lower values indicate better performance. Ego-vehicle frame: x = longitudinal, y = lateral. “Hold” denotes a naïve baseline that holds the last observed state; “Ours” denotes the proposed method.

Figure 6 illustrates the error distributions. The probability density of the terminal error at $+0.2s$ (Figure 6a) was sharply concentrated well below 1.0 meter, indicating that the model could localize the likely contact point with decimeter-to-submeter fidelity even under tight time constraints. The empirical cumulative distribution function (ECDF) curves (Figure 6c, 6d) further demonstrated the dominance of the proposed method, which maintained a steeper rise (i.e., a higher proportion of low-error cases) compared to all baselines across both horizons.

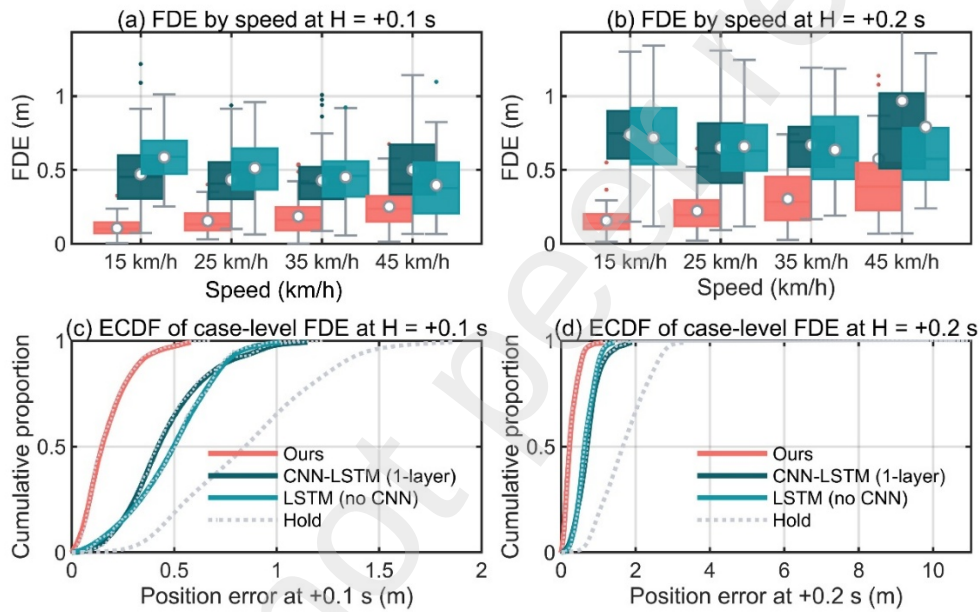


Fig. 6 Forecast accuracy and error distribution across models and horizons

3.3 End-to-end validation

The integrated framework was evaluated in end-to-end simulations across 28 stratified high-risk episodes. The prediction-based trigger policy achieved an overall correct decision rate of 92.9% (26/28 cases). Detailed per-case entries and the speed/decision breakdown are provided in Appendix A1. Specifically, the system exhibited a false activation rate of 7.1% (2/28), which occurred exclusively in high-speed non-collision scenarios ($v \geq 35$ km/h) involving late pedestrian withdrawals. A single missed activation (3.6%) was observed at 25 km/h, where the longitudinal closing speed derived from the forecast was slightly underestimated relative to the conservative trigger threshold.

Crucially, for the correctly identified collision cases, the proposed logic secured a pre-impact lead time of 200 ms (corresponding to the prediction horizon) compared to the reactive contact-triggered baseline. As illustrated in the system workflow (Figure 7), this anticipation provided the necessary time margin for the active hood to transition to its fully deployed state prior to head contact. Consequently, simulations confirmed that the deployed hood achieved a 62.1% reduction in the Head Injury Criterion (HIC) compared to the fixed hood baseline. This quantitative reduction verified that the sub-meter forecasting accuracy effectively enabled the protective device to function as intended within the high-risk time window.

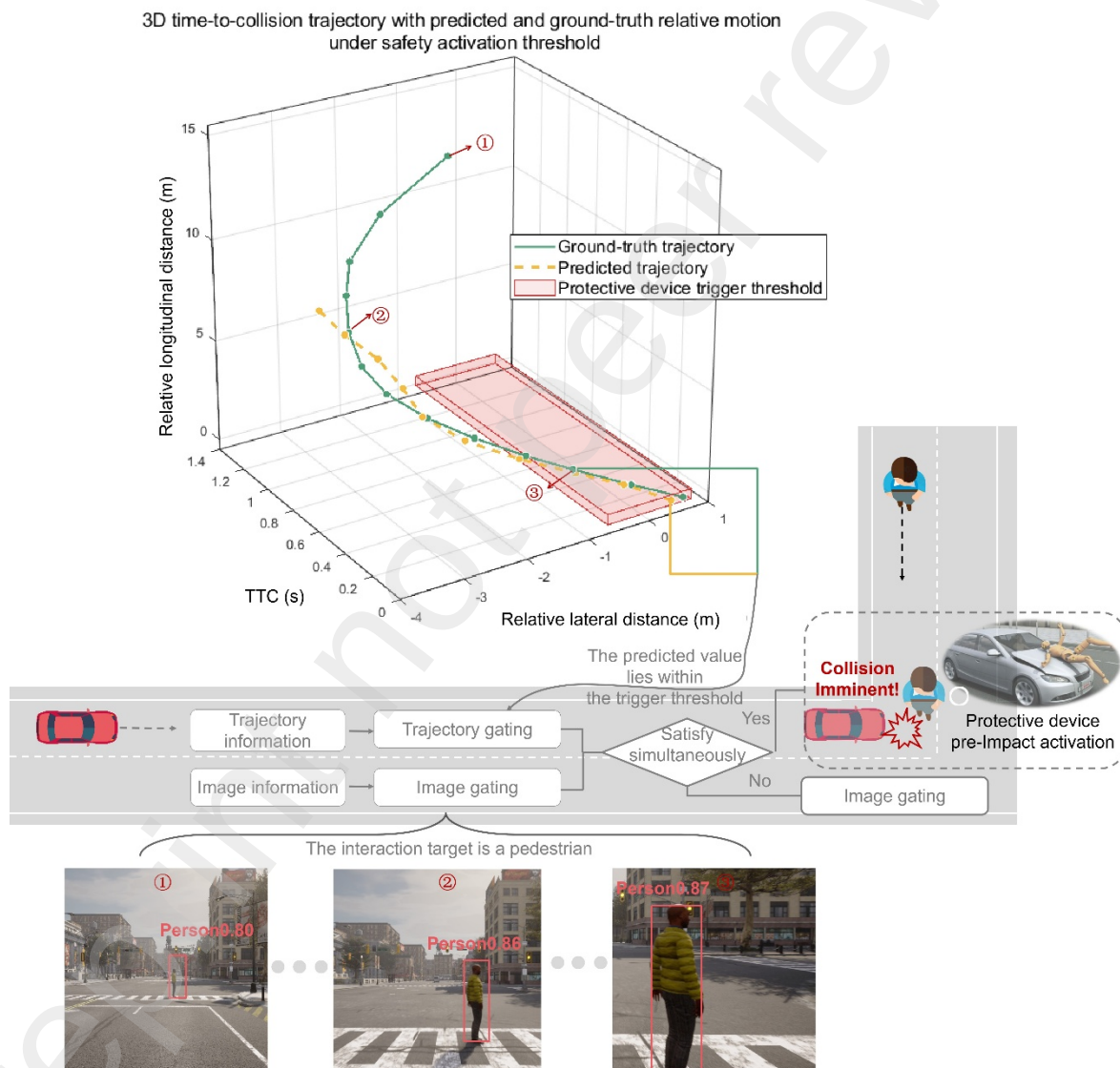


Fig. 7 End-to-end prediction-based trigger pedestrian protection workflow and timing benefit

Finally, the sim-to-real transferability was assessed using 65 real-world crash clips recorded by onboard sensors. When applied offline to these naturalistic sensor logs, the simulation-trained model maintained robust performance,

achieving a median FDE of 0.746 m and ADE of 0.676 m at the control horizon ($H = +0.2$ s) (Table 4). Notably, the model demonstrated superior lateral stability ($RMSE_y = 0.265$ m), which is critical for accurately gauging intrusion risk. Figure 8 visualizes these results in two aspects: the cumulative error curves (left panel) indicated that the proposed method consistently outperformed the baselines across the error spectrum; meanwhile, the representative case study (right panel) showed that the forecaster accurately locked onto the pedestrian's contact pose early in the final approach, even under dynamic lateral intrusion.

Table 4. Performance at $H = 0.20$ s for relative vehicle–pedestrian position trajectory prediction (real-case subset)

H (s)	Model	n (cases)	FDE, median [IQR] (m)	ADE, median [IQR] (m)	RMSE_x, mean (SD) (m)	RMSE_y, mean (SD) (m)
0.200	Ours	65	0.746 [0.503]	0.676 [0.538]	0.624 (0.368)	0.265 (0.190)
0.200	Shallow CNN+LSTM	65	0.869 [0.376]	0.809 [1.600]	1.543 (1.309)	0.464 (0.314)
0.200	LSTM-only	65	1.051 [0.913]	0.857 [0.948]	0.762 (0.302)	0.807 (0.601)
0.200	Hold	65	2.234 [0.919]	1.672 [0.694]	1.687 (0.548)	0.329 (0.419)

Notes: All rows use $n = 65$ cases. H = forecast horizon (seconds). FDE = final displacement error at H . ADE = average displacement error over $[0, H]$. $RMSE_x / RMSE_y$ = axis-wise root-mean-square error along vehicle longitudinal (x) and lateral (y) axes. IQR = interquartile range; SD = standard deviation. Units are meters; lower values indicate better performance. Ego-vehicle frame: x = longitudinal, y = lateral. “Hold” denotes a naïve baseline that holds the last observed state; “Ours” denotes the proposed method.

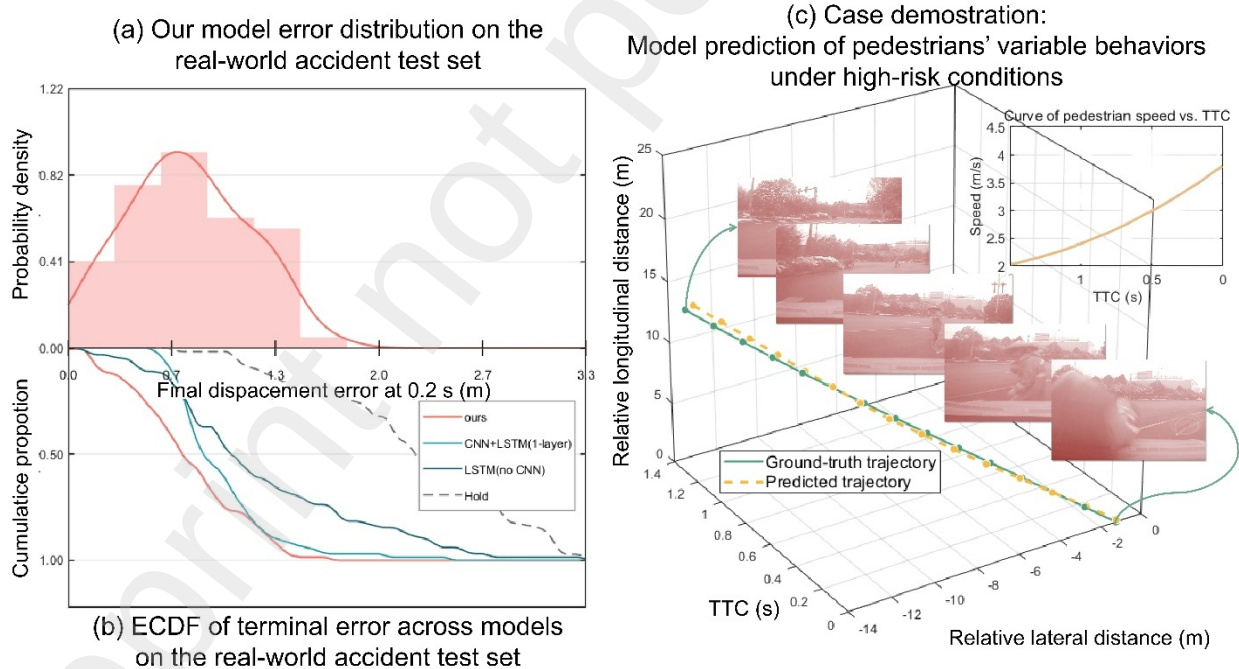


Fig. 8 Real-world deployment results: aggregate performance, baseline comparison, and pedestrian-evasive case study

4. Discussion

4.1 Efficacy and mechanism of pre-impact intervention

This study was designed to address a fundamental physical contradiction in pedestrian protection: reactive systems cannot overcome the mechanical latency of deployment within the shrinking time-to-collision of high-speed scenarios. Recognizing that long-term intent prediction is often unreliable in panic situations, our design philosophy prioritized short-horizon kinematic fidelity and strictly gated actuation. Integrated simulation results confirm the validity of this approach: by shifting the trigger point to a predicted "point of no return" (approx. 200 ms prior to impact), the framework effectively "buys back" the time lost to sensor processing. This temporal advantage enabled the active hood to fully deploy before head contact, directly translating into a 62.1% reduction in head injury severity relative to the fixed hood baseline.

This performance improvement is fundamentally attributed to addressing the data and model bottlenecks identified in the introduction. First, unlike standard datasets dominated by routine interactions, our HITL-CARLA corpus leverages the virtual environment to systematically generate "boundary cases"—near-misses and unavoidable collisions—while uniquely capturing pedestrians' active behaviors via the human-in-the-loop platform. This exposure allowed the model to learn subtle kinematic distinctions (e.g., aggressive gait changes) that define the collision boundary, overcoming the "average-case bias." Second, the causal CNN-LSTM architecture proved essential. The CNN stem stabilized noisy inputs into compact motion signatures, while the stacked LSTM captured nonlinear maneuver transitions. This architectural division of labor explains the consistent superiority of the proposed model over LSTM-only baselines (Table 3).

However, the failure modes observed reveal the dynamic limits of the current framework. The false activations at high speeds ($v \geq 35$ km/h) suggest that rapid changes in the longitudinal gap between sampling instants can induce transient prediction errors. Similarly, the missed activation at 25 km/h highlights the sensitivity of the speed-mapping function near the decision boundary. This indicates a fundamental trade-off between sensitivity (safety coverage) and specificity (nuisance avoidance) that must be carefully managed via gating logic.

4.2 Comparative safety benefits and framework generalizability

The significance of this framework is best understood through comparison with current industrial advancements. While emerging commercial systems (e.g., Tesla's vision-primed airbags) utilize pre-crash predictions primarily for occupant protection, our work extends this predictive paradigm to external vulnerable road users (VRUs). Unlike traditional active hoods that rely on contact sensors and thus fail to deploy in time during high-speed impacts due to mechanical latency, our framework demonstrates that pre-impact geometry—not just post-impact acceleration—can serve as a reliable trigger. This shift effectively bridges the gap between active collision avoidance and passive injury mitigation.

Beyond the specific demonstration with an active hood, the proposed framework offers broader engineering implications due to its device-agnostic nature. Most notably, by securing a stable 200 ms lead time, the framework relaxes the stringent hardware constraints that have historically limited pedestrian safety design. Traditionally, the need for millisecond-level response times forced reliance on non-reversible pyrotechnic actuators. Our prediction-based logic unlocks the feasibility of slower, reversible, but potentially more effective mechanisms—such as external pedestrian airbags or extendable bumpers—that were previously considered impractical for short-notice collisions. Furthermore, the explicit calibration of proximity gates and confidence thresholds provides an auditable safety template, facilitating integration into standardized functional safety assessments.

4.3 Limitations and future work

While the results demonstrate feasibility, several limitations define the current scope. First, the evidence base remains virtual. Although the human-in-the-loop protocol ensures behavioral realism, the domain gap between CARLA-rendered sensing and real-world environmental noise (e.g., glare, adverse weather) cannot be fully bridged by offline validation alone. Second, the current perception stack simplifies the aleatoric uncertainty inherent in real-world sensing, which is crucial for managing false-positive risks under heavy clutter. Finally, the injury assessment relied on MADYMO simulations at representative speeds; physical crash tests are ultimately required to validate the predicted HIC reductions in hardware.

These constraints naturally outline the roadmap for future work. The immediate next step is to transition from simulation to physical validation, utilizing closed-track proving grounds with safety envelopes to test end-to-end system latency and actuation reliability on production-grade hardware. In parallel, future iterations will incorporate

richer sensor fusion (e.g., radar, thermal) and uncertainty-gated control logic to enhance robustness. Ultimately, this prediction-based paradigm should be extended to enable a broader class of reversible countermeasures, thereby maximizing protection for vulnerable road users.

5. Conclusions

To address the critical reaction-time bottleneck in pedestrian protection systems under high-risk scenarios, this study proposed a prediction-based development framework that integrates a vehicle-view human-in-the-loop dataset, a short-horizon forecasting model, and a semantic-gated control policy. By systematically capturing "boundary cases"—such as near-misses and unavoidable collisions—the framework overcomes the limitations of traditional datasets that lack injury-oriented kinematic features.

Experimental results demonstrate that the proposed Causal CNN-LSTM model achieves robust sub-meter forecasting accuracy ($FDE = 0.220$ m at 0.2 s), effectively capturing abrupt pedestrian maneuvers that challenge traditional trajectory predictors. Crucially, the prediction-based trigger logic secures a stable 200 ms lead time relative to reactive contact-based systems. This temporal advantage enables the active hood to fully deploy prior to head contact, resulting in a 62.1% reduction in the Head Injury Criterion (HIC) in end-to-end simulations. Furthermore, validation on real-world accident cases confirmed the framework's robustness and transferability from virtual environments to naturalistic scenarios.

These findings highlight the importance of high-fidelity pre-impact prediction in mitigating injury risks when collision avoidance is physically infeasible. Ultimately, this work validates the feasibility of shifting the safety paradigm from post-impact detection to pre-impact prediction, providing a verifiable and device-agnostic pathway to unlock the next generation of integrated active and passive pedestrian safety systems.

Author statement

All authors have read and approved the final document.

Authorship contributions

Huamu Sun: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing - Original Draft, Writing - Reviewing and Editing. **Siyuan Liu:** Methodology, Validation, Investigation, Writing - Reviewing

and Editing. **Quan Li:** Methodology, Validation, Investigation, Writing - Reviewing and Editing. **Jiajie Shen:** Methodology, Software, Validation - Reviewing. **Xin Gao:** Methodology, Investigation, Visualization - Reviewing. **Xiangdong Ji:** Data curation, Methodology - Reviewing. **Yajun Zhang:** Data curation, Methodology - Reviewing. **Dongpeng Kou:** Data curation, Resources - Reviewing. **Bingbing Nie:** Conceptualization, Methodology, Writing - Reviewing and Editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix

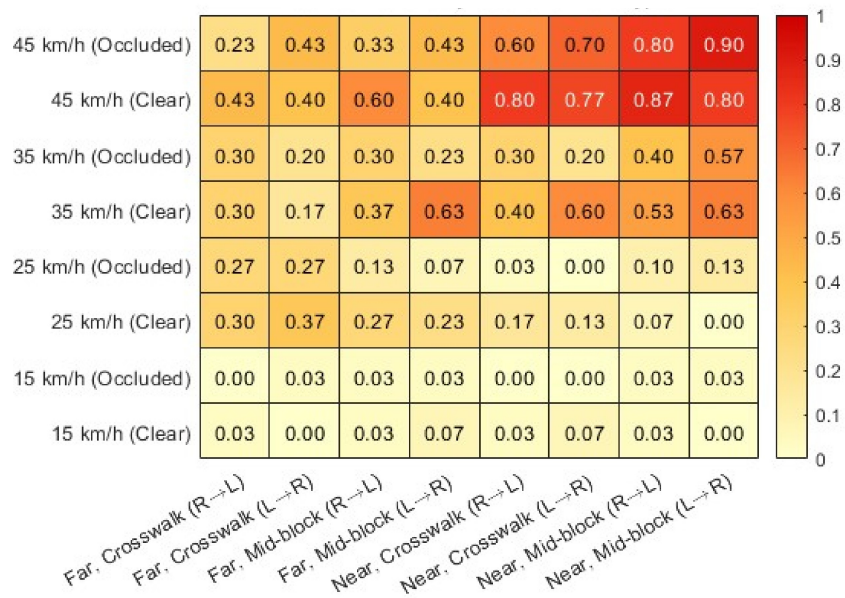
Appendix A. Validation details and supplementary results

Table A. 1. Stratified evaluation of prediction-based trigger hood decisioning

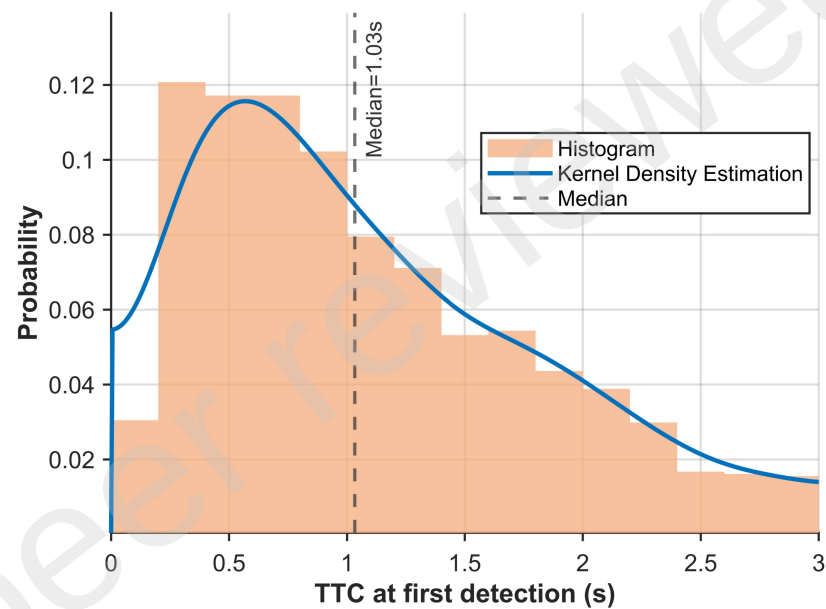
ID	Experiment ID	v [km/h]	Pedestrian decision	Actual collision?	Predicted collision?	Correct decision?
1	Ep018-01	15	Wait in place	No	No	Yes
2	Ep018-08	15	Accelerate forward	No	No	Yes
3	Ep019-12	15	Back off	No	No	Yes
4	Ep018-01	15	Normal walking	No	No	Yes
5	Ep008-23	25	Wait in place	No	No	Yes
6	Ep016-07	25	Accelerate forward	No	No	Yes
7	Ep020-16	25	Back off	No	No	Yes
8	Ep026-06	25	Normal walking	No	No	Yes
9	Ep020-05	25	Wait in place	Yes	Yes	Yes
10	Ep030-21	25	Accelerate forward	Yes	Yes	Yes
11	Ep020-48	25	Back off	Yes	Yes	No
12	Ep029-45	25	Normal walking	Yes	Yes	Yes
13	Ep019-35	35	Wait in place	No	No	Yes

14	Ep028-03	35	Accelerate forward	No	No	Yes
15	Ep027-51	35	Back off	No	Yes	No
16	Ep028-14	35	Normal walking	No	No	Yes
17	Ep016-59	35	Wait in place	Yes	Yes	Yes
18	Ep012-51	35	Accelerate forward	Yes	Yes	Yes
19	Ep028-39	35	Back off	Yes	Yes	Yes
20	Ep001-07	35	Normal walking	Yes	Yes	Yes
21	Ep025-26	45	Wait in place	No	No	Yes
22	Ep012-02	45	Accelerate forward	No	No	Yes
23	Ep024-58	45	Back off	No	Yes	No
24	Ep021-18	45	Normal walking	No	No	Yes
25	Ep025-34	45	Wait in place	Yes	Yes	Yes
26	Ep024-34	45	Accelerate forward	Yes	Yes	Yes
27	Ep027-34	45	Back off	Yes	Yes	Yes
28	Ep018-34	45	Normal walking	Yes	Yes	Yes

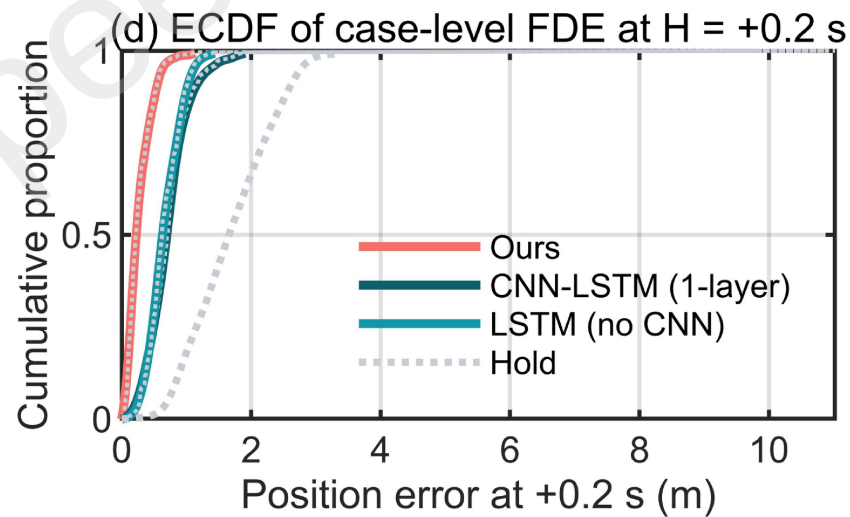
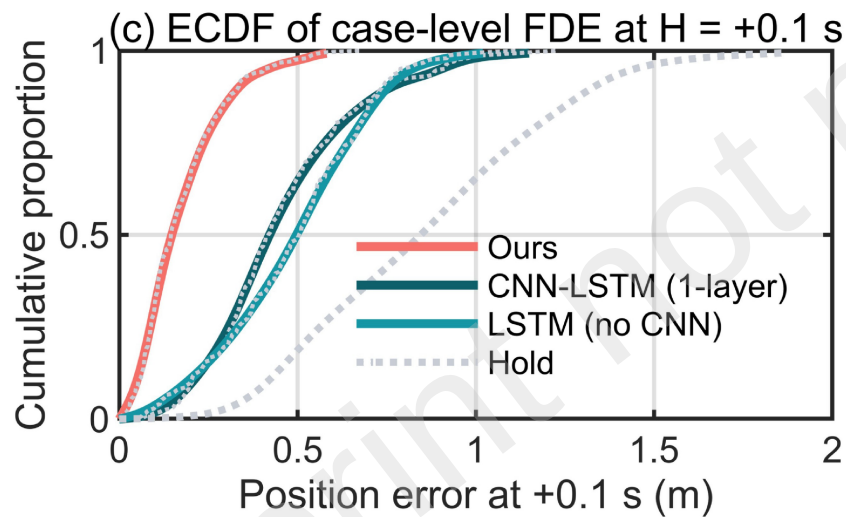
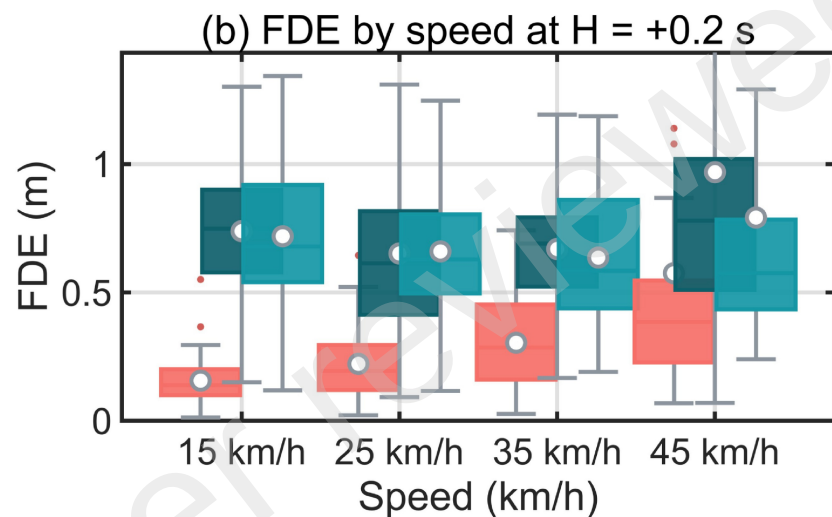
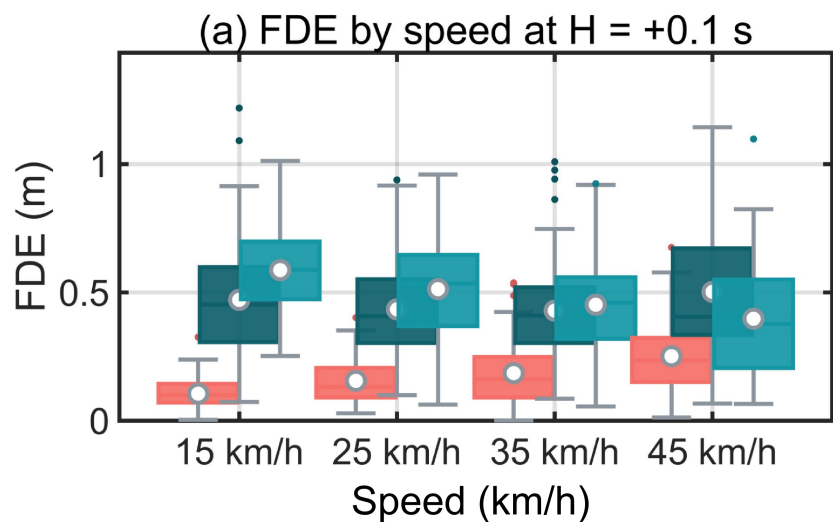
Notes: Planned n = 32 cases (stratification: 4 speeds × 4 pedestrian decisions × {collision, non-collision}); executed n = 28 because at 15 km/h no collision episodes were available, so only the four non-collision subcases were sampled. Condition-prediction outcomes: 26/28 correct (92.86%). Actuation outcomes (deployable hood): 2/28 false activations (7.14%) and 1/28 missed deployment (3.57%). The two false activations occurred at 35 km/h and 45 km/h under late back-off decisions (non-collision episodes classified as collision), and the missed deployment occurred at 25 km/h in a correctly predicted collision, attributable to conservative gating and speed-to-injury mapping ($|y| \leq 0.8$ m, $x < 0.5$ m; benchmark 25 km/h).

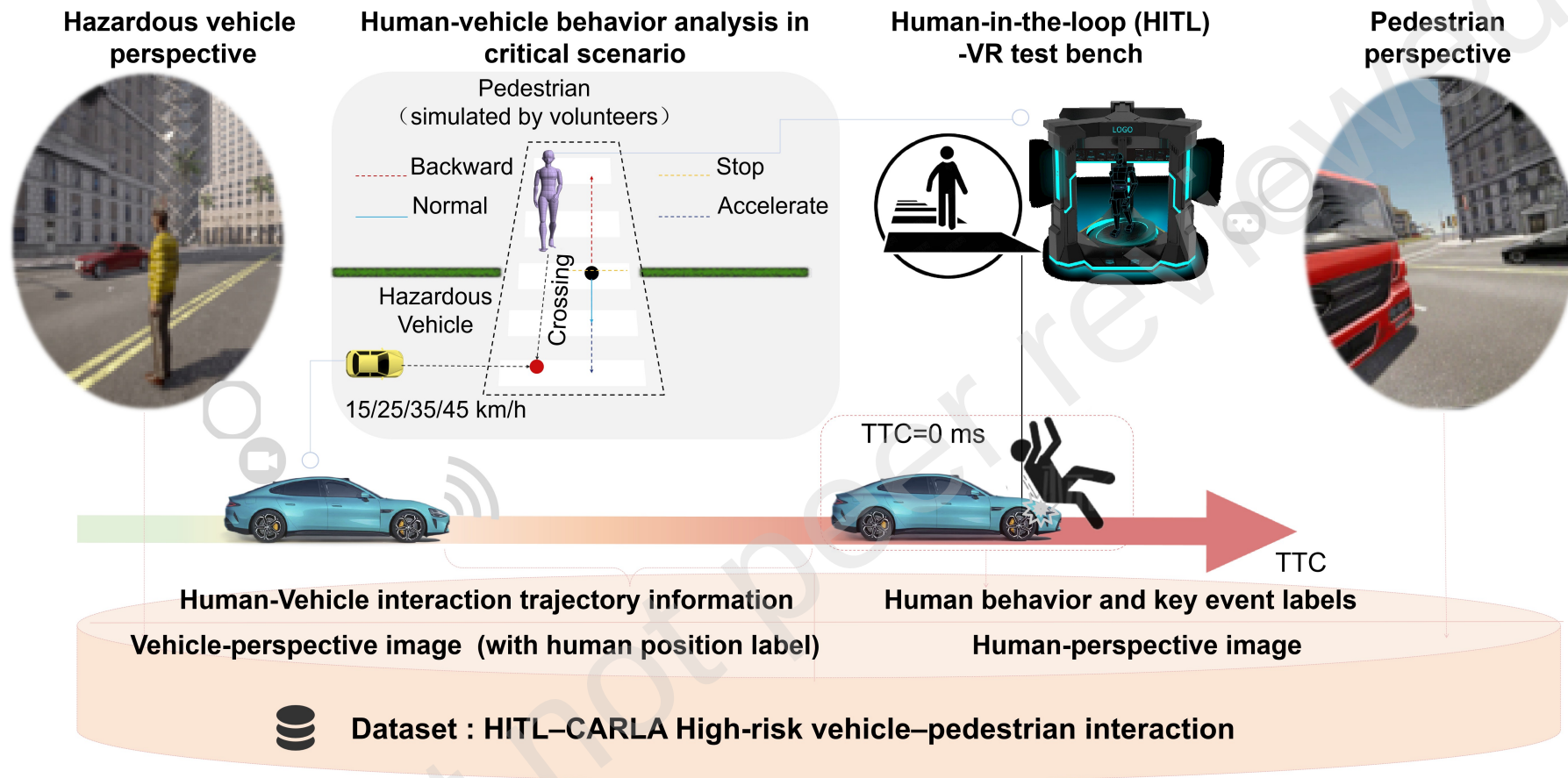


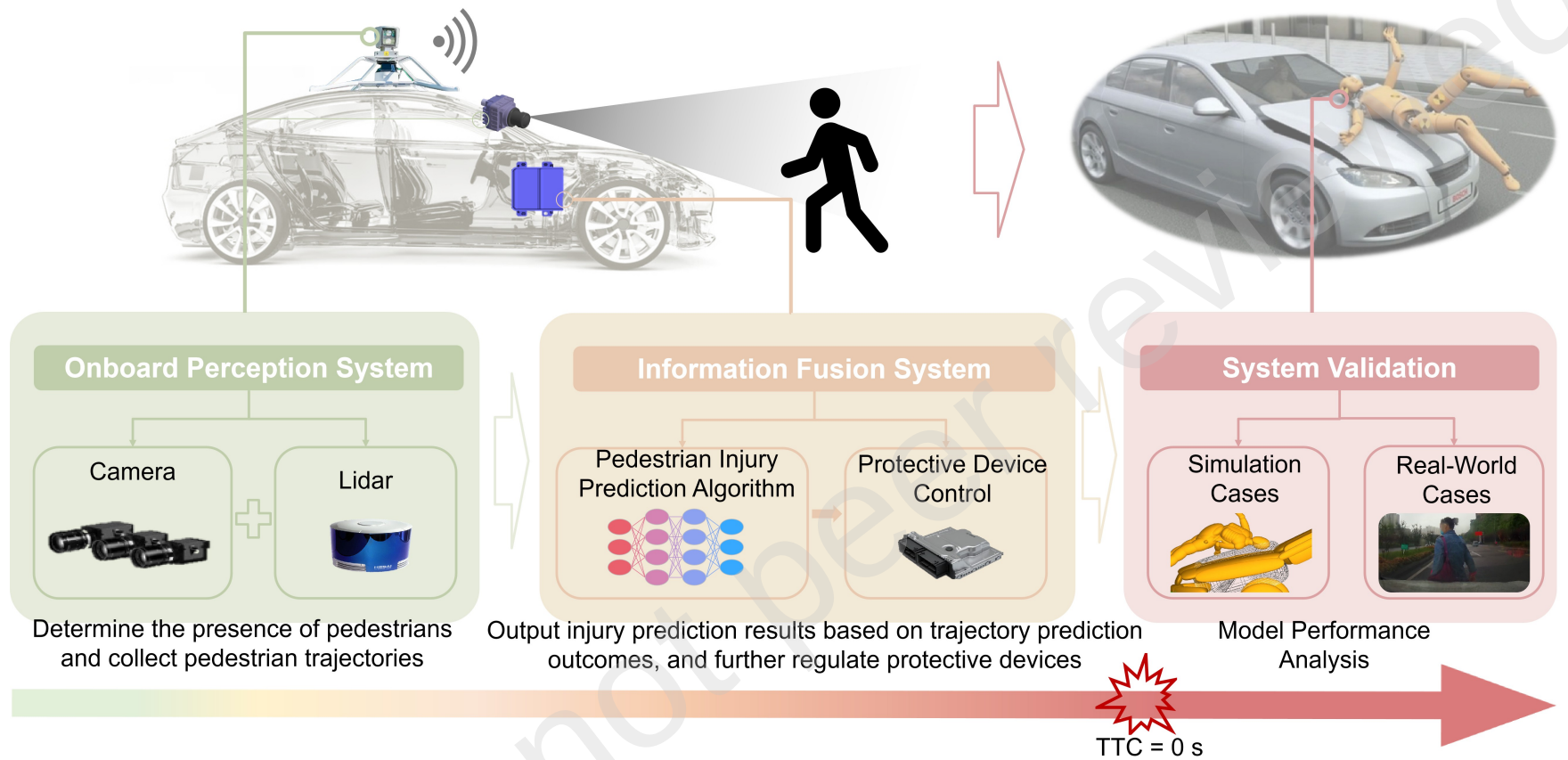
(a) Per-archetype collision share heatmap



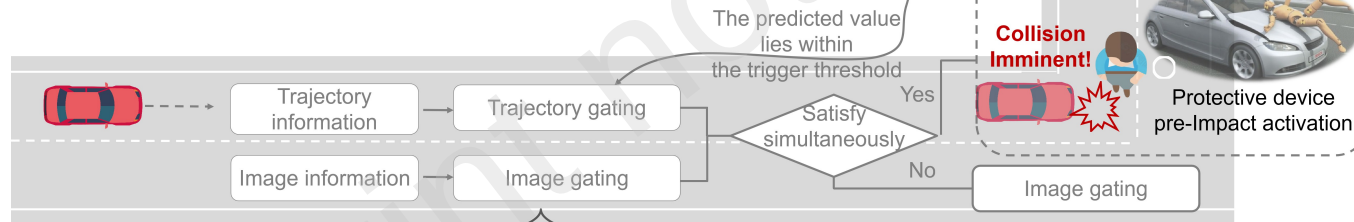
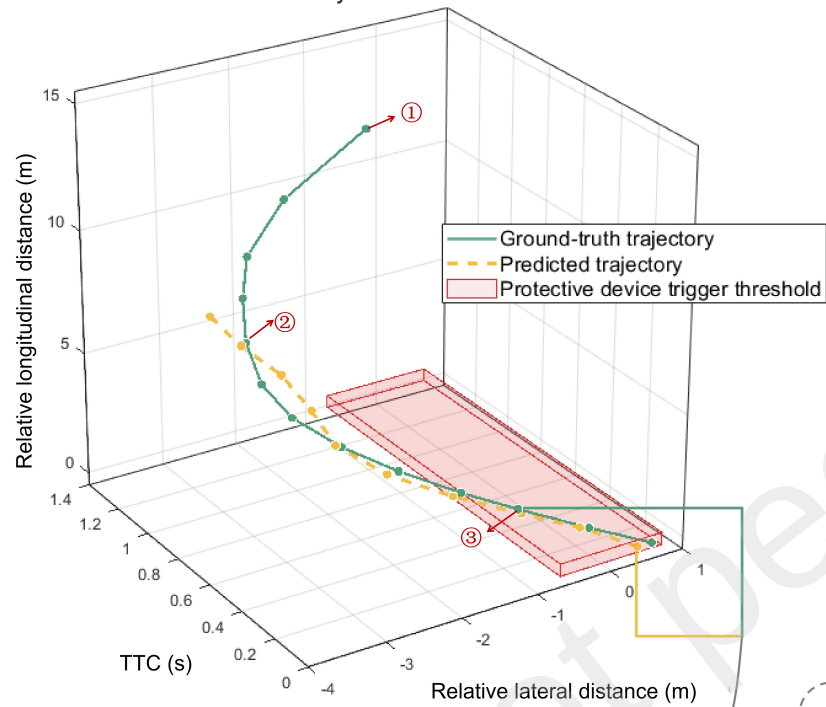
(b) Dataset-level TTC distribution



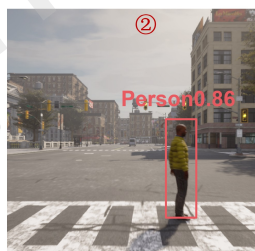




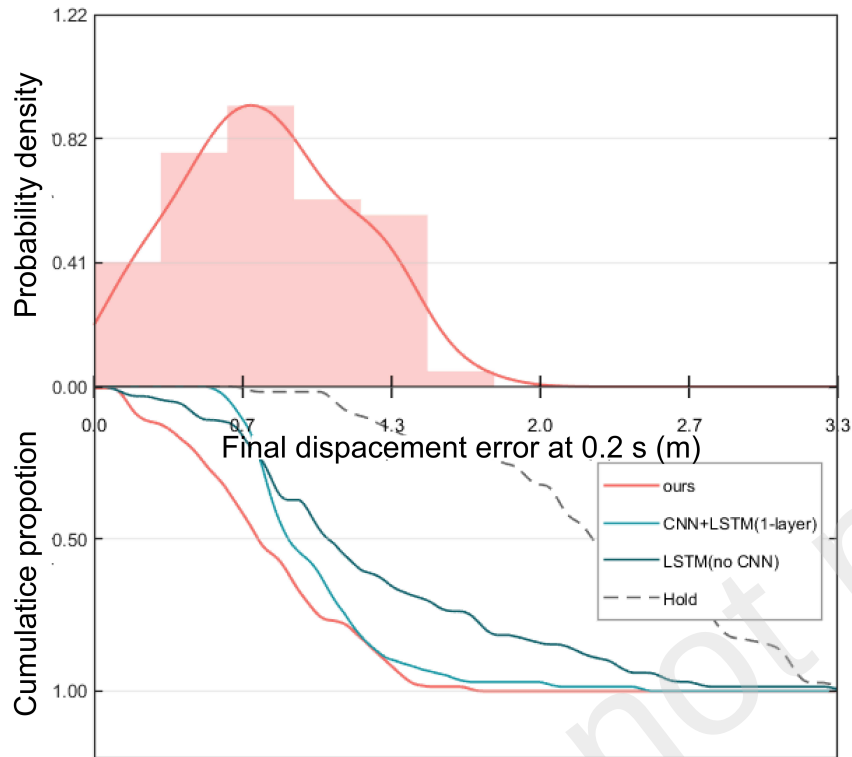
3D time-to-collision trajectory with predicted and ground-truth relative motion under safety activation threshold



The interaction target is a pedestrian



(a) Our model error distribution on the real-world accident test set



(b) ECDF of terminal error across models on the real-world accident test set

(c) Case demonstration:
Model prediction of pedestrians' variable behaviors under high-risk conditions

